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# A Yale Professor's Investment Formula Says You Need More Stocks. See How It Works.

The formula incorporates income, risk tolerance and other factors absent from typical rules of thumb

*By Joe Pinsker* Follow *and Peter Santilli* Follow
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Are you invested too much in stocks or not enough? There is a new way to answer that question.

Yale University finance professor James Choi recently developed a formula that recommends an asset allocation based in part on your age, income, savings and risk tolerance. The formula is drawn from a paper he co-authored last year and was adapted for The Wall Street Journal.

In many scenarios, the formula recommends a more aggressive, stock-heavy portfolio than other popular guidelines.

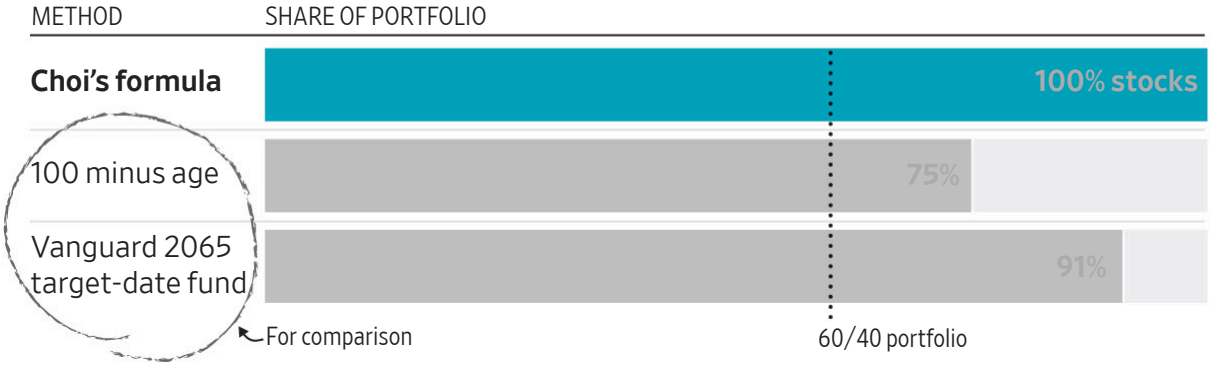
It also incorporates more of people's financial circumstances than common rules of thumb for equity allocation, such as the classic 60/40 division of stocks and bonds or subtracting your age from 100. Many simply outsource the decision to a target-date fund based on the year they plan to retire.

Take the example of a 25-year-old who makes $70,000 a year after taxes and has $25,000 to invest.

Example 1 of 3

**A young worker**

| 25 years old | $70k annual income | $25k investable net worth |
|---|---|---|




Sources: Choi et al.; Vanguard

The formula's central insight is that the future paychecks and retirement benefits that someone has yet to receive in their life are, when taken as a whole, like a bond because fluctuations in earnings aren't strongly correlated with stock-market returns. For this 25-year-old, that big, bondlike chunk of future money means they could more easily weather a steep drop in stocks.

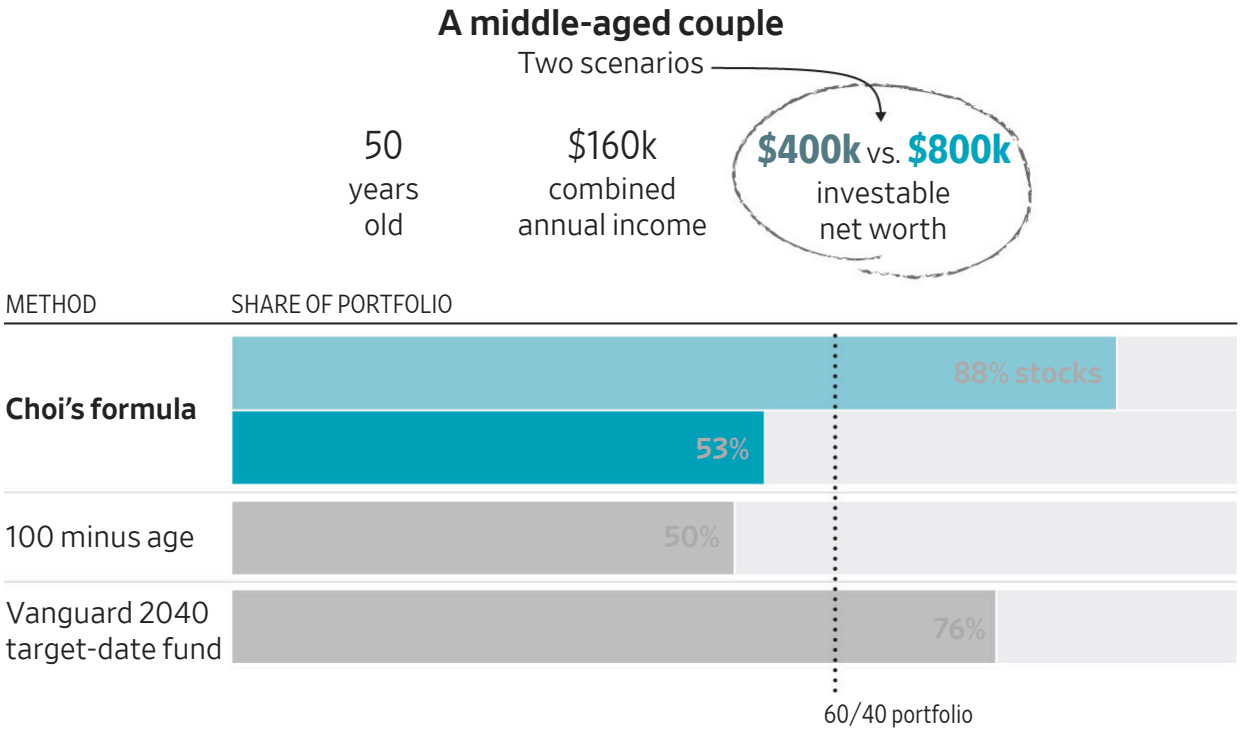
"The market dropped 20%, but you still have another 30 years of wages coming to you, so you can actually absorb that loss—no problem," Choi said.

The formula comes from a paper Choi released last year with Pengcheng Liu and Canyao Liu, current and former Ph.D. students of his, respectively. Their paper turned decades-old academic insights about future labor earnings into something everyday investors can use.

The math is complicated, so Choi recently made a spreadsheet, intended as an educational tool, where people can plug in their financial specifics, tweak the assumptions and see what the formula spits out.

As another example, the way it treats a hypothetical middle-aged couple differs based on the amount they have saved up.

Example 2 of 3




In both scenarios, the formula steers the couple toward a lower equity allocation than it did the 25-year-old, because a smaller share of their lifetime income is still to come.

But it recommends a much lower allocation, 53%, when the couple has twice as much money to invest, because in that scenario upping the equity allocation would alter the risk profile of a larger proportion of their projected lifetime resources.

"It's more conservative when you have more money saved up," Choi said.

The formula doesn't aim to maximize the amount of money investors accumulate, but rather the "utility" they get from spending their money over the course of a lifetime, since that utility declines with each successive dollar.

The researchers calculate in their paper that for a hypothetical 22-year-old investor, the formula on average leads to 0.06% less utility from lifetime spending in a variety of situations than if the investor's portfolio were allocated optimally. By comparison, the "100 minus your age" rule of thumb yields 2% less utility than that perfect portfolio.

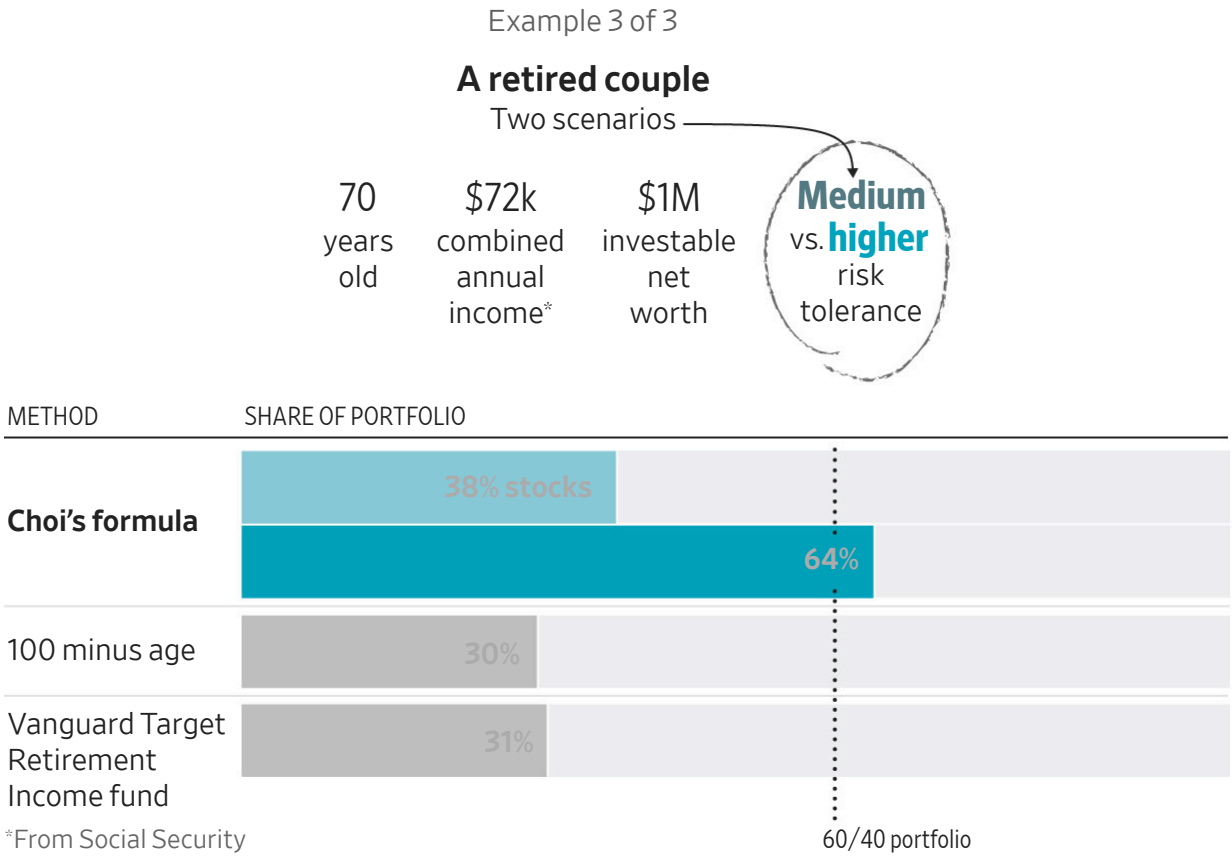
That is a difference, but not an enormous one. Choi noted that the amount of money someone saves up to invest in the first place is generally more important to their financial well-being than how they allocate those investments.

Economist Vicki Bogan, who wasn't involved with creating the formula, views it as a useful bridge between academic theory and everyday investors' needs.

"It might not be perfect, but it gives a solid, kind of directionally correct way to think about your portfolio choice," said Bogan, a professor at Duke University's Sanford School of Public Policy.

Another key input to the formula is an investor's appetite for putting more money on the line to reap bigger gains. It captures that with a score for risk aversion from one to 10 (with 10 being the most averse to risk), which people can determine for themselves by answering a few thought-experiment questions.

A higher or lower tolerance for risk can make a significant difference, as seen in the case of an older couple when their score is a more cavalier 3, versus 5, out of 10.




"For some people, the upside matters more and for some people, the loss hurts more," said Choi.

These examples assume that investors' income growth matches an average college graduate's and that their year-to-year chances of dying match government mortality data. The formula also incorporates an investor's expectation for the stock market's average annual inflation-adjusted return.

The uncertainty of forecasting one's future income, stock-market performance and other variables is an important caveat of using the formula, said Terrance Odean, a finance professor at UC Berkeley's Haas business school.

Choi recognizes this caveat but said he thinks it is valuable to incorporate even a rough approximation of, say, future earnings rather than exclude them from financial planning.

He also mentioned the limitation that in its current form, the formula doesn't account for people's home equity or mortgage debt, though neither do many other investing guidelines.

For a previous research project, Choi read dozens of popular personal-finance books and tracked the investing advice he encountered most.

"Even relatively equity-heavy advice that you see commonly is, for my taste, probably not aggressive enough," he said.

Joe Pinsker is a reporter for The Wall Street Journal's personal finance section. He covers the pursuit of happiness—how people feel about money and the economy, how they're defining a good life, and the obstacles and avenues to wealth and prosperity in America....

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Peter's graphic-based stories and interactives have covered a broad array of subjects within finance, including Tesl...